FOR A FOUR-TERM INVERSE SYSTEM OF COMPLEXES THE ZEROTH COHOMOLOGY OF THE LIMIT IS THE LIMIT OF THE ZEROTH COHOMOLOGIES UNDER TWO VANISHING HYPOTHESES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The counted statement span contains the author's four-term display, delivered verbatim. The counted proof span contains the author's footnote on injectivity, including its 'Details omitted.', and two displayed pages written as matrices; all are delivered verbatim. The author writes 'A inverse system' in the footnote, reproduced unchanged. The single reference inside the counted proof span is recorded as a non-load-bearing navigation pointer.

Allowed standard prerequisites (editorial).

- The right derived functor R lim on D(Ab(N)) and the first derived limit R^1 lim. used throughout the counted proof span. Supplies the terms appearing in both spectral sequences.
- The two spectral sequences computing the cohomology of a complex of complexes under a functor. invoked at proof lines 23044-23050. Are the two computations whose comparison yields the conclusion.
- Injectivity in the category of inverse systems of abelian groups. explained in the footnote at proof lines 23045-23049. Legitimises the use of the spectral sequences by providing enough injectives.

Source reference. Original lemma 015J: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1510–1510. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let*

$$(A_n^{-2} \to A_n^{-1} \to A_n^0 \to A_n^1)$$

be an inverse system of complexes of abelian groups and denote $A^{-2} \to A^{-1} \to A^0 \to A^1$ *its limit. Denote* (H_n^{-1}), (H_n^0) *the inverse systems of cohomologies, and denote* H^{-1}, H^0 *the cohomologies of* $A^{-2} \to A^{-1} \to A^0 \to A^1$. *If*

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) (A_n^{-2}) *and* (A_n^{-1}) *have vanishing* $R^1\lim$,
(2) (H_n^{-1}) *has vanishing* $R^1\lim$,

then $H^0 = \lim H_n^0$.

Proof. Let $K \in D(Ab(\mathbf{N}))$ be the object represented by the system of complexes whose nth constituent is the complex $A_n^{-2} \to A_n^{-1} \to A_n^0 \to A_n^1$. We will compute $H^0(R\lim K)$ using both spectral sequences[1] of Derived Categories, Lemma 015J. The first has E_1-page

$$\begin{array}{cccc} 0 & 0 & R^1\lim A_n^0 & R^1\lim A_n^1 \\ A^{-2} & A^{-1} & A^0 & A^1 \end{array}$$

with horizontal differentials and all higher differentials are zero. The second has E_2 page

$$\begin{array}{cccc} R^1\lim H_n^{-2} & 0 & R^1\lim H_n^0 & R^1\lim H_n^1 \\ \lim H_n^{-2} & \lim H_n^{-1} & \lim H_n^0 & \lim H_n^1 \end{array}$$

and degenerates at this point. The result follows. □

[1]To use these spectral sequences we have to show that $Ab(\mathbf{N})$ has enough injectives. A inverse system (I_n) of abelian groups is injective if and only if each I_n is an injective abelian group and the transition maps are split surjections. Every system embeds in one of these. Details omitted.